

Model Answer - Grade 9 Physical Science - Forces and Motion - Short Test

Q1: C. With zero net force an object keeps its velocity (Newton's first law).

Q2: B. The newton (N) is the unit of force.

Q3: C. $F = m \times a = 2 \text{ kg} \times 3 \text{ m/s squared} = 6 \text{ N}$.

Q4: Two forces act on the book: its weight (gravity), downward, and the normal force from the table, upward. They are equal in size and opposite in direction, so the net force on the book is zero. With zero net force there is no acceleration.

Q5a: B. At a steady speed the forward force equals the total resistive force.

Q5b: The cyclist's speed is not changing, so the acceleration is zero and the net force is zero. The forward force from the pedals therefore balances the resistive forces (air resistance and friction).